Privacy-Preserving Big Data System using Data Masking Techniques

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1. Objective

- To design a Big Data system that preserves user privacy using data masking techniques.
- To protect sensitive information such as personal details, financial data, and health records.
- To enable secure data analytics without exposing confidential information.

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2. Learning Outcomes

After completing this practical, students will be able to:

- Understand privacy challenges in Big Data environments.
- Learn different data masking techniques for sensitive data protection.
- Implement privacy-preserving data processing workflows.
- Analyze masked datasets without compromising data confidentiality.
- Gain knowledge of secure data sharing and compliance requirements.

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3. Software and Hardware Requirements

Software

- Python / Java
- Big Data framework (Apache Hadoop / Apache Spark)
- Jupyter Notebook / Google Colab
- Libraries: Pandas, NumPy, PySpark

Hardware

- Minimum 8GB RAM (recommended for Big Data processing)
- Intel i5 or higher processor
- 20GB storage

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4. Theory

4.1 Big Data Privacy

Big Data systems collect and process massive volumes of user information. Protecting sensitive data is essential to prevent identity theft, data breaches, and unauthorized access.

4.2 Privacy-Preserving Techniques

- Data masking
- Encryption
- Anonymization
- Access control

4.3 Data Masking

Data masking hides original sensitive values while maintaining data usability.

Types of Data Masking:

- **Static masking:** Masked copy of stored data
- **Dynamic masking:** Data masked during access
- **Tokenization:** Replace data with tokens
- **Shuffling:** Rearranging values within dataset
- **Substitution:** Replace data with realistic fake values

4.4 Importance

- Prevent data leakage
- Ensure regulatory compliance
- Allow safe data analytics

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5. System Architecture

```
        Raw Sensitive Big Data
    (PII, Financial, Health Data)
                  |
                  ▼
          Data Ingestion Layer
                  |
                  ▼
         Data Masking Module
(Tokenization, Substitution, Shuffling)
                  |
                  ▼
      Secure Storage / Big Data Cluster
                  |
                  ▼
       Analytics & Visualization Layer
                  |
                  ▼
          Privacy-Preserving Insights
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6. Procedure (Step-by-Step)

Step 1: Import libraries and initialize Big Data environment

- Load Pandas / PySpark and required tools.

Step 2: Load dataset

- Import dataset containing sensitive attributes (Name, Email, Phone, Salary).

Step 3: Identify sensitive fields

- Detect personally identifiable information (PII).

Step 4: Apply data masking techniques

- Replace names with pseudonyms
- Mask phone numbers and email IDs
- Shuffle sensitive numerical values

Step 5: Store masked dataset

- Save masked data into distributed storage.

Step 6: Perform analytics

- Run queries and analysis on masked dataset.

Step 7: Display results

- Compare original vs masked dataset.

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7. Output (Example)

Name	Email	Phone	Salary
A***	a***@mail.com	XXXXX1234	50,000
B***	b***@mail.com	XXXXX5678	65,000

Sensitive information is successfully hidden.

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8. Result

The system successfully masked sensitive information while allowing analytics on large datasets, ensuring privacy preservation and secure data processing.

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9. Applications

- Healthcare data analytics
- Banking and financial systems
- Government databases
- E-commerce platforms
- Cloud data sharing

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10. Advantages

- Prevents data breaches
- Maintains data usability
- Enables secure data sharing
- Supports regulatory compliance

- Protects user identity

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11. Conclusion

This practical demonstrates how data masking can be integrated into Big Data systems to preserve privacy while maintaining analytical value. The approach ensures secure processing of sensitive information and is essential for modern data-driven organizations.

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12. Viva Questions

1. What is data masking and why is it important in Big Data?
2. Difference between static and dynamic data masking?
3. How does tokenization help in privacy preservation?
4. Can analytics be performed effectively on masked data?